

★ What is WebElement in Selenium?

WebElement is an **interface** in Selenium that represents a **single HTML element** on a web page. Once Selenium finds an element using a **locator**, it returns a **WebElement object**, and then we perform actions on it.

```
WebElement username = driver.findElement(By.id("user-name"));
```

findElement() → finds the element

username → WebElement

Actions are performed using **WebElement methods**

★ Why WebElement is Important?

- ✓ To enter text. - send keys
- ✓ To click buttons. - click
- ✓ To read text - getText
- ✓ To check element status (displayed, enabled, selected)
- ✓ To get attributes, CSS values, size, location
- 👉 Without WebElement, we cannot interact with UI elements

Common WebElement Methods

◆ sendKeys()

- Used to enter text
- Works on input fields and text areas

Use cases:

- Login forms
- Search boxes
- File upload

◆ clear()

- Clears existing text from input fields
- Used before re-entering data

◆ click()

- Clicks buttons, links, checkboxes, radio buttons

◆ getText()

- Returns **visible text** of an element
- Used for validations

Important:

- Does NOT work for input field values

◆ getAttribute()

- Fetches attribute value
- Used to read input box values

◆ isDisplayed()

- Checks if element is visible on UI
- Returns boolean

Use cases:

- Logo validation
- Element presence check

◆ isEnabled()

- Checks if element is enabled or disabled
- Mostly used for buttons

◆ isSelected()

- Used for:
 - Checkbox
 - Radio button
 - Dropdown options

Returns:

- true → selected
- false → not selected